MIL-T-17585C(YD)
14 February 1983
SUPERSEDING
MIL-T-17585B(YD)
18 June 1974

MILITARY SPECIFICATION

THERMOMETERS, RECORDING, CAPILLARY TUBE AND BULB, AND RESISTANCE

This specification is approved for use by the Naval Facilities Engineering Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers capillary tube and bulb (filled), and resistance (electrical) automatic recording thermometers.

1.2 Classification. Recording thermometers shall be of the following types and sizes, as specified (see 6.2).

Type I - Capillary tube and bulb.
Type II - Resistance.
Size 4R - 4-inch nominal diameter chart.
Size 6R - 6-inch nominal diameter chart.
Size 8R - 8-inch nominal diameter chart.
Size 10R - 10-inch nominal diameter chart.
Size 12R - 12-inch nominal diameter chart.
Size 9S - Strip chart - 9-inches or more nominal width.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. Unless otherwise specified (see 6.2), the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

PPP-B-636 - Boxes, Shipping, Fiberboard.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer (Code 156), Naval Construction Battalion Center, Port Hueneme, CA 93043, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 6685

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MILITARY

MIL-P-116 - Preservation, Methods of.
MIL-V-173 - Varnish, Moisture- and Fungus-Resistant, (For the Treatment of Communications, Electronic, and Associated Equipment).

STANDARDS

FEDERAL

FED-STD-H28 - Screw-Thread Standards for Federal Services.

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.
MIL-STD-147 - Palletized Unit Loads.
MIL-STD-794 - Parts and Equipment, Procedures for Packaging and Packing of.

(Copies of specifications and standards, required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. The issue of the document which is indicated as DoD adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

E644 - Industrial Resistance Thermometers, testing.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

No. 70 - National Electrical Code.

(Application for copies should be addressed to the National Fire Protection Association, Batterymarch Park, Quincy, MA 02269.)

(Industry association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Description.

3.1.1 Type I, capillary tube and bulb. A type I recording thermometer shall consist of a temperature sensing and measuring (filled) system, a

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recording system, a chart-drive mechanism, a case, and shall use size 4R, 6R, 8R, 10R, or 12R charts, as specified (see 6.2). Unless otherwise specified (see 6.2), 100 charts shall be furnished.

3.1.2 Type II, resistance. A type II recording thermometer shall consist of a measuring system, a balancing system, a recording and indicating system, a chart-drive mechanism, a case, and shall use size 10R, 12R, or 9S charts, as specified (see 6.2). Unless otherwise specified (see 6.2), 100 charts shall be furnished.

3.2 First article. When specified (see 6.2), the contractor shall furnish a complete thermometer for first article inspection and approval (see 4.2.1 and 6.4).

3.3 Standard commercial product. The thermometers shall, as a minimum, be in accordance with the requirements of this specification and shall be the manufacturer's standard commercial product. Additional or better features which are not specifically prohibited by this specification but which are a part of the manufacturer's standard commercial product, shall be included in the thermometers being furnished. A standard commercial product is a product which has been sold or is being currently offered for sale on the commercial market through advertisements or manufacturer's catalogs, or brochures, and represents the latest production model.

3.4 Materials. Materials used shall be free from defects which would adversely affect the performance or maintainability of individual components or of the overall assembly. Materials not specified herein shall be of the same quality used for the intended purpose in commercial practice. Unless otherwise specified herein, all equipment, material, and articles incorporated in the work covered by this specification are to be new and fabricated using materials produced from recovered materials to the maximum extent possible without jeopardizing the intended use. The term "recovered materials" means materials which have been collected or recovered from solid waste and reprocessed to become a source of raw materials, as opposed to virgin raw materials. None of the above shall be interpreted to mean that the use of used or rebuilt products are allowed under this specification unless otherwise specified.

3.5 Master lock and key. When specified (see 6.2), the manufacturer's standard master lock and key shall be furnished.

3.6 Components.

3.6.1 Type I thermometers.

3.6.1.1 Sensing and measuring system. The sensing bulb, a capillary tube, and either a bourdon tube, metallic bellows, metallic diaphragm capsules, or a nonmetallic (elastomer) diaphragm shall constitute a hermetically sealed system for the containment of the measuring fluid. The materials of this system and the materials used in joining these components and in sealing the system shall not react chemically with each other or with the measuring fluid. Interior surfaces shall be smooth and free from burrs, beads, oxides, or other imperfections which could adversely affect the functioning of the system.

3.6.1.2 Bulb. The bulb shall be either averaging, plain with union connection or with separable socket connections, as specified (see 6.2). The bulb or, when specified, the separable socket shall be suitable for exposure to the medium, pressure, and the temperature conditions, as specified (see 6.2).

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3.6.1.3 Capillary tubing. Capillary tubing shall be of a uniform cross section throughout its length and protected by spirally wound armor of bronze, corrosion-resistant steel or armor covered with a jacket of polyvinyl chloride or lead covered, as specified (see 6.2). When full compensation is used, the compensating means and the primary capillary tubing shall be encased within the same armored protecting tube. When specified (see 6.2), flexible armored tubing at the case shall be reinforced by a 3-inch length of spirally wound armor or an equivalent reinforcing means. A sleeve at the end of the reinforced, spirally wound armor shall prevent uncoiling.

3.6.1.4 Temperature sensors. The temperature sensors shall have a high resistance to drift and progressive set. The temperature-sensitive bulb shall be capable of occasional exposure to a temperature 10 percent of range span above the upper range limit with no harmful effect on performance. The temperature sensors shall be capable of the endurance test (see 4.6.3) without exhibiting more than 0.5 percent drift in calibration.

3.6.1.5 Recording system. The movement of the recording system shall be free, smooth, and capable of producing a substantially uniform trace throughout the calibrated span. A means shall be provided for making positive and accurate adjustments to the zero-position of the tracing point and for minor recalibration. The penpoint shall have longtime, free-flow characteristics and shall not abrade the chart. The penpoint, or the penpoint and arm assembly when not separable, shall be readily removable for cleaning. A pen lifter, which holds the arm in the lifted position until manually released, shall be provided so that the pen arm can be lifted away from the chart. Operation of the lifter shall not cause ink to spatter. When the pen is in its highest lifted position, the ink shall not run along the pen arm or into the recorder. When in the released position, the pen lifter shall not interfere with the free movement of the pen arm or the chart. The pen, pen arm, and connecting linkage shall be of corrosion-resistant materials. The linkage and the pen arm shall be sufficiently rigid to withstand normal handling during servicing without suffering any permanent deformation.

3.6.1.6 Chart drive. The chart shall be driven by a hand-wound spring motor or by an electric motor, as specified (see 6.2).

3.6.1.7 Spring motor. The spring motor shall operate continuously at the specified speed for 7 days between windings and shall be enclosed in a dustproof case. The movement shall be of corrosion-resistant material. The motor shall be equipped with a speed regulator which can be easily adjusted from the outside of the clockcase. Unless the motor is self-starting, a means shall be provided for starting. The winding key shall be integral with or separable from the motors. When the key is separable, a keyholder shall be provided within the case. The winding direction, the speed adjusting directions, and the starting directions shall be clearly indicated by a permanent legend adjacent to each knob or lever.

3.6.1.8 Electric motor. The electric motor shall be of the self-starting, synchronous type equipped with permanently lubricated, antifriction bearings. The enclosure shall be dustproof.

3.6.1.9 Chart mount. The chart mount shall be such that each chart will be automatically, accurately centered and will lie flat against the chart plate. A means shall be provided to keep the chart locked on its mount so as to eliminate relative motion between the chart and its shaft. The locking

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means shall be permanently assembled with or fastened to the thermometer, so that it cannot be lost. The chart plate shall be firmly held in place, but shall be readily removable. A means shall be provided for keeping the outer edge of the chart close to the chart plate without binding, wrinkling, or tearing the chart. A means, such as an index, shall be provided for setting the chart at the proper time line. Special tools shall not be required to replace charts.

3.6.1.10 Case. The case for round chart thermometers shall be round or rectangular for flush or wall mounting, as specified (see 6.2). It shall be dustproof and equipped with a hinged door held firmly closed by an integral latch or lock to cause compression of a sponge rubber, molded rubber, or similar door seal material affixed to either the door or the case. The door shall be equipped with a gasketed, or otherwise sealed, clear glass observation window that permits a full view of the chart. A stop or check shall be provided to limit the amount of door opening. The entrance for the capillary tube shall be at the back or bottom, as specified (see 6.2). The capillary tubing shall be anchored at the point of entrance in a manner to avoid strain on the operating mechanism. The anchorage shall be of a type that permits ready removal of the bourdon tube. The case shall be cast-aluminum alloy or sheet steel or other material used in commercial practice.

3.6.2 Type II thermometers. Type II thermometers shall be the self-balancing or direct deflection type.

3.6.2.1 Measuring system. The measuring system of a type II thermometer shall measure temperature converted into either a change in electrical resistance or voltage. Changes in the temperature (see 3.7.3) of the sensing element shall cause the measuring system to produce a signal that causes operation of the balancing system. The measuring system shall have longtime stability of electrical and magnetic characteristics, which shall not be impaired by temperature, stray fields, or vibration. Components shall be protected against electrical and mechanical damage from dust and moisture. Sliding contact parts shall be nontarnishing, have constant and low contact resistance, and give long-life contact with negligible wear.

3.6.2.2 Balancing system. The balancing system shall respond, without appreciable lag, to the signal of the measuring system and shall operate in the direction which causes the unbalanced signal to diminish, so that the accuracy shall be within the specified limits (see 3.7.2). The movement shall be positive, smooth, and free of appreciable backlash or lost motion.

3.6.2.3 Electronic amplifiers. Electronic amplifiers shall be of a solid-state design. Amplifier assemblies shall be readily removable as a unit. A readily removable fuse or other means shall be provided to protect the amplifier against electrical damage. A readily accessible means shall be provided for adjusting the sensitivity. The power input and signal input terminal blocks or sockets shall be separated as widely as possible. Small parts, such as resistors or capacitors, mounted by their leads shall be so mounted as to remain in proper position and not be subject to strain because of thermal expansion or contraction.

3.6.2.4 Recording system. The recording system shall be capable of producing a definite, clear, and substantially uniform record on the chart. The lateral position and the intensity of pressure of the recording point, stamp, or die against the chart shall be adjustable.

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3.6.2.5 Continuous-record recorders. The penpoints of continuous-record recorders shall have longtime, free-flow characteristics and shall not abrade the chart. The penpoint, or the penpoint and pen-support assembly when not separable, shall be readily removable for cleaning.

3.6.2.6 Intermittent-record recorders. The arrangement shall be such that the printing sequence and the temperature-element switching sequence will be in step at all times. The switch shall make contact with only one temperature-element circuit at a time. Ink pads and printing stamps or dyes shall be readily accessible for inspection and service. Ink pads shall be readily replaceable.

3.6.2.7 Chart-drive mechanism. The chart-drive mechanism shall cause the chart to move uniformly at the specified rate without wrinkling or tearing the chart or causing lateral drift. The chart-drive motor shall be of the self-starting synchronous type. A toggle switch shall be provided for stopping the chart drive without de-energizing the other parts of the equipment. The chart mount shall be of the type that permits the roll of paper to be put in place and threaded to the reroll quickly and without special tools. When specified (see 6.2), a tear-off device shall be provided.

3.6.2.8 Case. The case shall be rectangular for flush or wall mounting, as specified (see 6.2), and be equipped with a hinged door held firmly closed by an integral latch or lock to cause compression of a sponge rubber, molded rubber, or similar door seal material affixed to either the door or to the case. The door shall be equipped with a gasketed or otherwise sealed, clear glass, observation window that permits a full view of the chart, scale indicator, and scale. A stop or check shall be provided to limit the amount of door opening. The case shall be cast-aluminum alloy or sheet steel or other material used in commercial practice.

3.6.2.9 Indicator and scale. The indicator and scale shall provide easy readability at a distance. Numerals, scale subdivisions, and index shall be black, and the scale background shall be white.

3.6.2.10 Sensing elements. When specified (see 6.2), sensing elements with fittings (separable sockets, bushings, flanges, etc.) shall be furnished. Elements shall be suitable for the electrical characteristics and calibration on the instrument for which they are furnished. When specified (see 6.2), leads for connecting the sensing element to the thermometer shall be furnished.

3.7 Performance. The thermometer shall measure and record temperatures within the temperature range, as specified (see 6.2), at locations which are at the specified distance (length of capillary tubing or leads, see 6.2) from the thermometer. The number of locations served by a type I thermometer shall be one, two, three, or four, as specified (see 6.2). The number of locations served by a type II thermometer shall be from 1 to 16 inclusive, as specified (see 6.2). The temperature shall be recorded on a paper chart of the size specified (see 3.1.1, 3.1.2, and 6.2). Marking shall be accomplished by use of an ink or an inkless writing system. The record shall be either a continuous line or a series of symbols, such as dots, dashes, or crosses. When more than one record is made on a single chart, each record shall be clearly identifiable, either by the use of differently colored inks (up to and including six records) or by the printing of a number or a letter adjacent to each printed symbol.

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3.7.1 Chart scales. When the temperature range specified for type I thermometers has its lower limit above -40° Fahrenheit (F) and its higher limit below +400°F, the temperature scale on the chart shall be either uniformly graduated or the graduations shall become progressively larger from the center of the chart outward as the temperature scale increases, as specified (see 6.2). When the temperature limits are above +400°F, the temperature scale on the chart shall be uniformly graduated. The temperature scale for type II thermometers shall be uniformly graduated.

3.7.1.1 Scale range. The specified temperature range shall encompass approximately 80 percent or more of the temperature scale printed on the chart.

3.7.1.2 Scale division and numbering. The chart shall be divided and numbered as shown in table I.

TABLE I. Scale division and numbering.

Chart scale range (degrees)	Division numbering (degrees)	Subdivision graduation (degrees)
100	10	1
200	20	2
500	50	5
1000	100	10
2000	200	20

3.7.2 Sensitivity. Whenever the temperature change recorded by the type II thermometer is more than 0.2 percent of the specified temperature span, the recorder pen shall reposition in the direction of the change (see 3.6.2.1 and 4.7).

3.7.3 Accuracy. For type I thermometers, instrument accuracy shall not vary ±0.5 percent with tubing and case at ambient temperature as specified (see 6.2), and tubing length is within manufacturer's limitations. Ambient error is ±0.5 percent for ±25°F ambient temperature change. For type II thermometers, instrument accuracy (for instrument only) shall be ±0.5 percent of the span (see 3.6.2.2).

3.7.4 Chart speeds. Round charts shall make one revolution in 24 hours or one revolution in 7 days, as specified (see 6.2). Strip charts shall move at a rate of 1-, 2-, or 3-inches per hour for single-record recorders, as specified (see 6.2), and at a rate of 2-, 4-, or 6-inches per hour for multi-record recorders, as specified (see 6.2).

3.7.5 Pen-traverse speed. After actuation, the recording element shall be capable of traversing 80 percent of the full scale of the instrument in not more than 15 seconds for type I thermometers, not more than 30 seconds for single-record type II thermometers, and not more than 45 seconds for multi-record type II thermometers (see 4.6.2 and 4.7).

3.7.6 Printing interval. The normal interval between successive printed points shall be not more than 30 seconds. When the records are so spaced that in one interval of the complete cycle the recording carriage must travel for 30 seconds or more, the interval shall not exceed 1 minute.

3.7.7 Ambient temperature and humidity. The thermometer shall be capable of continuous operation after a period of idleness in any ambient temperature

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between the ambient temperature limits specified (see 6.2), and in any relative humidity of 80 percent or less, without sustaining mechanical or electrical damage. Except for vapor pressure systems, automatic compensation shall be provided for changes in temperature at the case.

3.8 Electrical equipment. All electrical equipment shall operate from a 115 volt (V), 60 Hertz power supply. Variations in voltage at the instrument between 105 V and 125 V shall not cause faulty operation. Enclosures of electrical and mechanical instruments having electrical devices must be grounded in accordance with the rules and regulations of article 250 National Electrical Code, NFPA No. 70.

3.9 Accessibility. All components housed within the thermometer case shall be completely accessible from the front. Parts and subassemblies shall be so mounted that their removal shall be feasible without damage to adjacent parts. Parts most likely to require inspection, service, or replacement shall be so located that minimum disassembly is required.

3.10 Vibration. The equipment shall be sufficiently rugged to withstand the vibration normally encountered in industrial-type use without damage or impairment of function.

3.11 Interchangeability. All thermometers of the same classification furnished with similar options under a specific contract shall be identical to the extent necessary to insure interchageability of component parts, assemblies, accessories, and spare parts.

3.12 Threads. Screw threads for all threads securing or attaching devices and threaded parts shall be in accordance with FED-STD-H28 or as follows:

a. Machine threads shall be the standard national coarse or fine thread series.
b. Pipe threads shall be the standard National taper pipe thread series.

3.13 Instruction plates. The thermometer shall be equipped with instruction plates suitably located, describing any special or important procedures to be followed in operating and servicing the equipment. Plates shall be of a material which will last and remain legible for the life of the equipment and shall be securely affixed.

3.14 Identification marking. Identification shall be permanently and legibly marked directly on the thermometer case or on a corrosion-resisting metal plate securely attached to the thermometer at the source of manufacturer. Identification shall include the manufacturer's model and serial number, name and trademark to be readily identifiable to the manufacturer.

3.15 Treatment and painting Unless otherwise specified (see 6.2), the thermometers shall be treated and painted in accordance with the manufacturer's standard practice. All surfaces of the thermometer other than corrosion-resisting steel shall be protected against corrosion and present a neat appearance.

3.16 Fungus resistance. When specified (see 6.2), electrical components and circuit elements, including terminal and circuit connections, shall be coated with varnish conforming to MIL-V-173, except that:

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a. Components and elements inherently inert to fungi or in hermetically sealed enclosures need not be coated.
b. Current-carrying contact surfaces, such as relay contact points, shall not be coated.

3.17 Workmanship.

3.17.1 Steel fabrication. The steel used in fabrication shall be free from kinks, sharp bends, and other conditions which would be deleterious to the finished product. Manufacturing processes shall not reduce the strength of the steel to a value less than intended by the design. Manufacturing processes shall be done neatly and accurately. All bends shall be made by controlled means to insure uniformity of size and shape.

3.17.2 Bolted connections. Boltholes shall be accurately punched or drilled and shall have the burrs removed. Washers or lockwashers shall be provided in accordance with good commercial practice, and all bolts, nuts, and screws shall be tight.

3.17.3 Riveted connections. Rivet holes shall be accurately punched or drilled and shall have the burrs removed. Rivets shall be driven with pressure tools and shall completely fill the holes. Rivet heads, when not countersunk or flattened, shall be of approved shape and of uniform size for the same diameter of rivet. Rivet heads shall be full, neatly made, concentric with the rivet holes, and in full contact with the surface of the member.

3.17.4 Welding. Welding procedures shall be in accordance with a nationally recognized welding code. The surface of parts to be welded shall be free from rust, scale, paint, grease, or other foreign matter. Welds shall be of sufficient size and shape to develop the full strength of the parts connected by the welds. Welds shall transmit stress without permanent deformation or failure when the parts connected by the weld are subjected to proof and service loadings.

3.17.5 Castings. All castings shall be sound and free from patching, misplaced coring, warping, or any other defect which reduces the castings ability to perform its intended function.

3.17.6 Machine work. Tolerances for contact and bearing surfaces shall conform to standards prevailing among manufacturers normally producing recording thermometers.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Component and material inspection. Components and materials shall be inspected in accordance with all the requirements specified herein and in applicable referenced documents.

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4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.2.1).
b. Quality conformance inspection (see 4.2.2).

4.2.1 First article inspection. The first article inspection shall be performed on one thermometer of each classification specified when a first article is required (see 3.2 and 6.4). This inspection shall include the examination of 4.5 and the tests of 4.6. The first article may be either a first production item or a standard production item from the supplier's current inventory provided the item meets the requirements of the specification and is representative of the design, construction, and manufacturing technique applicable to the remaining items to be furnished under the contract.

4.2.2 Quality conformance inspection. Quality conformance inspection shall be performed on the sample thermometers selected in accordance with 4.4. This inspection shall include the examination of 4.5, the tests of 4.6 and the packaging inspection of 4.9.

4.3 Inspection lot. All thermometers of the same classification offered to the Government at one time shall be considered a lot for purposes of inspection. The sample unit shall be one complete thermometer.

4.4 Sampling. Sampling and inspection procedures shall be in accordance with MIL-STD-105. All thermometers offered for delivery at one time shall be considered a lot for the purpose of inspection. If an inspection lot is rejected, the contractor may rework it to correct the defects, or screen out the defective units, and resubmit for a complete reinspection. Resubmitted lots shall be reinspected using tightened inspection. If the rejected lot was screened, reinspection shall be limited to the defect causing rejection. If the lots reprocessed, reinspection shall be performed for all defects. Rejected lots shall be separate from new lots, and shall be clearly identified as reinspected lots.

4.4.1 Sampling for examination. Examination shall be based on inspection level II and an Acceptable Quality Level (AQL) of 2.5 percent defective.

4.4.2 Sampling for tests. Tests shall be based on inspection level S-2 and an AQL of 2.5 percent defective.

4.5 Examination. Each sample selected shall be examined for defects listed in table II.

TABLE II. Classification of defects.

Classification	Defects	Requirement paragraph
Critical:	None defined.	
Major:		
101	Type 1 thermometer sensing and measuring system leaks fluid.	3.6.1.1
102	Bulb and capillary tubing not as specified.	3.6.1.2/3.6.1.3
103	Operation of the pen lifters causes ink splatter.	3.6.1.5

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TABLE II. Classification of defects - Continued.

Classification	Defects	Requirement paragraph
104	Pen arm movement not free and smooth.	3.6.1.5/3.6.2.5
105	No speed regulator adjustment.	3.6.1.7
106	Electric motor and enclosure not as specified.	3.6.1.8
107	Chart binds, wrinkles or tears.	3.6.1.9
108	Case not as specified.	3.6.1.10/3.6.2.8
109	Chart-drive toggle switch missing.	3.6.2.7
110	Indicator and scale not as specified.	3.6.2.9
111	Sensing elements and leads not as specified.	3.6.2.10
112	Chart graduations, range and scale division not in accordance with table I.	3.7.1/3.7.1.1/ 3.7.1.2
113	Chart speed and printing interval not as specified.	3.7.4/3.7.6
114	Voltage not as specified.	3.8
115	Component parts not accessible.	3.9
116	Component parts and accessories not interchangeable.	3.11
Minor:		
201	Number charts furnished for type I and type II thermometers not as specified.	3.1.1/3.1.2
203	Spring motor, winding, key and keyholder missing.	3.6.1.7
204	Type II thermometer amplifier has no protection against electrical damage.	3.6.2.3
205	Instruction plate not as specified.	3.13
206	Identification marking not as specified.	3.14

4.6 Tests. Each thermometer selected shall be tested in accordance with ASTM E644 and any thermometer failing to pass the tests of 4.6.1 through 4.8, as applicable, shall be rejected.

4.6.1 Calibration tests.

4.6.1.1 Type I. The sensitive portion of the sensing element, without separable socket, shall be inserted into a vessel containing a suitable liquid in such a way that the temperature of the sensing element and the bulb of the calibrating thermometer will be the same. The recorder and capillary tube shall be at normal room temperature (approximately 70°F) during this test. Calibration curves shall be obtained over the full range of the instrument with temperature ascending and descending at such a rate that the effect of lag will be eliminated. Failure to conform with 3.7.2 and 3.7.3 shall be cause for rejection.

4.6.1.2 Type II. Temperatures simulated by electrical quantities shall be impressed on the recorder. These shall be obtained from high-precision equipment and shall be checked frequently. The test shall be conducted by starting at the lowest point on the thermometer scale and increasing the values in steps equal to 10 percent of the range of the thermometer until the maximum simulated temperature is reached, then decreasing the values in steps equal to 10 percent of the range until the lowest point on the thermometer scale is reached. Failure to conform with 3.7, 3.7.2, and 3.7.3 shall be cause for rejection.

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4.6.2 Pen-traverse speed test.

4.6.2.1 Type I. Two agitated liquid baths, one having a temperature not lower than 95 percent of the upper range limit and the other having a temperature not higher than 5 percent of the lower range limit, shall be used. The sensitive element shall be immersed in the hot bath for approximately 10 minutes and then in the cold bath until the recorder pen is near the lowest point of the range. The sensitive element shall then be quickly transferred to the hot bath. Timing shall start when the recorder pen reaches 10 percent of full range and shall stop when the recorder pen reaches 90 percent of full range. Failure to conform to 3.7.5 shall be cause for rejection.

4.6.2.2 Type II. A simulated temperature at approximately the bottom of the range shall be impressed upon the recorder after it has been in operation 5 minutes. Then the voltage shall be increased to a value corresponding to a simulated temperature at approximately the top of the range. Timing shall start when the recording point reaches 10 percent of full range and shall stop when the recording point reaches 90 percent of full range. The test shall first be made with the electrical power supply at 115 V, then repeated at 105 V, and at 125 V. Failure to conform to 3.7.5 shall be cause for rejection.

4.6.3 Endurance test. The thermometer shall be subjected to 300 full cycles of operation (see 3.6.1.4). The sensing elements of type I instruments shall be dipped alternately in a hot and cold bath or equivalent means shall be employed to produce travel of the recorder pen of approximately 80 percent of the range. Type II thermometers shall have impressed upon them simulated temperatures which will cause alternate upscale and downscale movement of the recording mechanism, approximately 80 percent of the range in each direction. Mechanical or electrical damage to the thermometers shall be cause for rejection.

4.7 Performance test. All thermometers shall be adjusted and individually calibrated at three points within the range of the thermometer. The calibration points shall be in the lower third, the middle third, and the upper third of the range. During calibration each instrument shall be tested for pen-traverse speed (see 3.7.5). Each type II thermometer shall be tested for sensitivity (see 3.7.3). The tests shall be conducted at room temperature (approximately 70°F).

4.8 Vibration test. Visual examination shall show no defects. A shaker shall be used that is capable of producing sinusioidal motion in the acceleration ranges.

4.9 Packaging inspection. The preservation, packing, palletization, and marking of the item shall be inspected to verify conformance to the requirements of section 5.

5. PACKAGING

5.1 Preservation. Preservation shall be level A or commercial as specified (see 6.2).

5.1.1 Level A. Each recording thermometer shall be packaged method III in accordance with MIL-P-116 in a fiberboard box conforming to PPP-B-636, class weather-resistant and waterproof sealed in accordance with the appendix thereof.

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5.1.2 Commercial. The equipment shall be preserved in accordance with the contractor's standard practice in a manner to prevent deterioration and damage.

5.2 Packing. Packing shall be level A, B, or commercial as specified (see 6.2).

5.2.1 Levels A and B. Packing shall be in accordance with MIL-STD-794. Containers shall be selected from table I for the appropriate level.

5.2.2 Commercial. The complete equipment shall be packed in a manner which will insure arrival at destination in satisfactory condition. Preparation for delivery shall comply with applicable carrier rules and regulations.

5.3 Palletization. Material shall be palletized in accordance with MIL-STD-147 when the following criteria is met:

a. Load to consist of four or more unskidded containers; and,
b. Load shall utilize a minimum of 80 percent of the pallet base.

5.4 Marking. Marking shall be in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. Recording thermometers covered by this specification are intended for use in measuring and recording temperatures in the range between -350°F and +2800°F. Either type I or type II is suitable for use in measuring temperatures up to +1000°F. In the range above 1000°F, type II should be specified. Type I should not be selected when the length of capillary required is greater than 200 feet. When the capillary length is greater than 25 feet for a type I thermometer, the bulb size should be increased from that used with shorter capillaries to compensate for ambient temperature effects on the capillary itself. When a type I thermometer is to be used with charts having graduations which become progressively larger as the temperature increases (see 3.7.1), it is essential that the elevation of the sensing bulb in relation to the recorder case be specified within the nearest foot.

6.2 Ordering data. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. Type and size of thermometers and charts desired (see 1.2).
c. Issue of document required if other than as specified (see 2.1.1).
d. Number and size of charts required for type I or type II thermometers (see 3.1.1 and 3.1.2).
e. When a first article is required for inspection and approval (see 3.2, 4.2.1, and 6.4).
f. When the manufacturer's standard master lock and key is required (see 3.5).
g. Type of bulb required, pertinent recording thermometer systems dimensions medium in which bulb is immersed and pressure of the medium for type I thermometers (see 3.6.1.2).
h. Type of capillary tubing required and reinforcement at the case, when as specified (see 3.6.1.3).
i. When chart drive for type I thermometers is spring wound or electric motor driven (see 3.6.1.6).

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j. Type of case required, round or rectangular, and type of mounting. Location of entrance for the capillary tubing is as specified (see 3.6.1.10).
k. When a tear-off device for type II chart mounts is required (see 3.6.2.7).
l. Type of case mounting required for type II thermometers (see 3.6.2.8).
m. Type of sensing elements and whether leads are required for type II thermometers (see 3.6.2.10).
n. Range of temperature to be measured, length of capillary tubing or leads, and number of locations served by type I or type II thermometers (see 3.7).
o. Chart scale graduations for type I thermometers desired between -40°F and +400°F, as specified (3.7.1).
p. Ambient temperature limits for capillary tubing used for type I thermometers, as specified (3.7.3).
q. Round chart speed and strip chart speeds for single and multi-records, as specified (see 3.7.4).
r. Ambient temperature and humidity limits as specified (see 3.7.7).
s. When treatment and painting is other than as specified (see 3.15).
t. When fungus resistance for electrical components is required (see 3.16).
u. Level of preservation and level of packing required (see 5.1 and 5.2).

6.3 Data requirements. When this specification is used in an acquisition which incorporates a Contract Data Requirements List (DD Form 1423), and invokes the provisions of paragraph 7-104.9(n) of the Defense Acquisition Regulations (DAR), the data requirements will be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the approved DD Form 1423 incorporated into the contract. When the provisions of DAR 7-104.9(n) are not invoked, the data shall be delivered in accordance with the contract requirements.

6.4 First article. When a first article inspection is required, the item will be tested and should be a first production item or it may be a standard production item from the contractor's current inventory as specified in 4.2.1. The first article should consist of one thermometer. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examination, test, and approval of the first article.

6.5 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

Review activity:

DLA - GS

Preparing activity:

Navy - YD

Project No. 6685-N552

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